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Koehler

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(54) **FLEXING CHAIR SHELL WITH FLEXIBLE ELEMENTS**

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A47C 5/12	(2006.01)
A47C 7/44	(2006.01)

(52) **U.S. Cl.**

CPC *A47C 7/445* (2013.01); *A47C 5/12* (2013.01); *A47C 3/12* (2013.01)

(58) **Field of Classification Search**

CPC A47C 3/12; A47C 5/12; A47C 7/445
See application file for complete search history.

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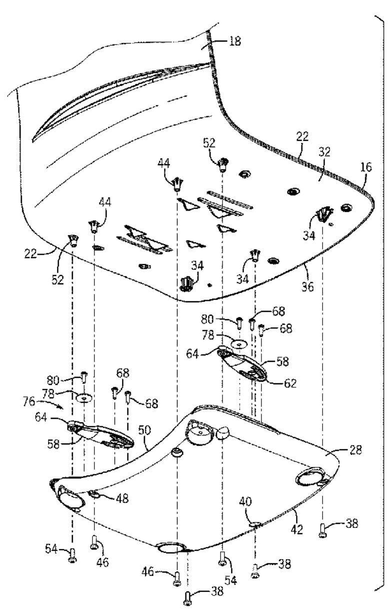
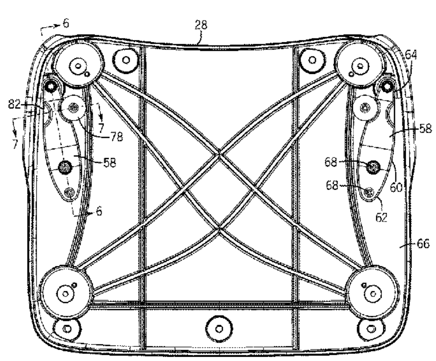
(57) **ABSTRACT**

A chair having a one-piece molded shell including a seat and a backrest that can recline relative to the seat. One or more flexible elements are mounted to the seat such that the flexible elements restrict the inward flexing of the seat as the backrest reclines. The flexible elements are positioned to allow the initial flexing of the side edges of the seat as the backrest reclines relative to the seat. A stop member is positioned relative to the flexible elements such that the stop member limits the amount of movement of the flexible elements. The flexible elements are located between the seat and the base. The flexible elements exert a bias force on the side edges of the seat to urge the backrest to return to an initial position when a reclining force is not longer exerted on the backrest.

17 Claims, 9 Drawing Sheets





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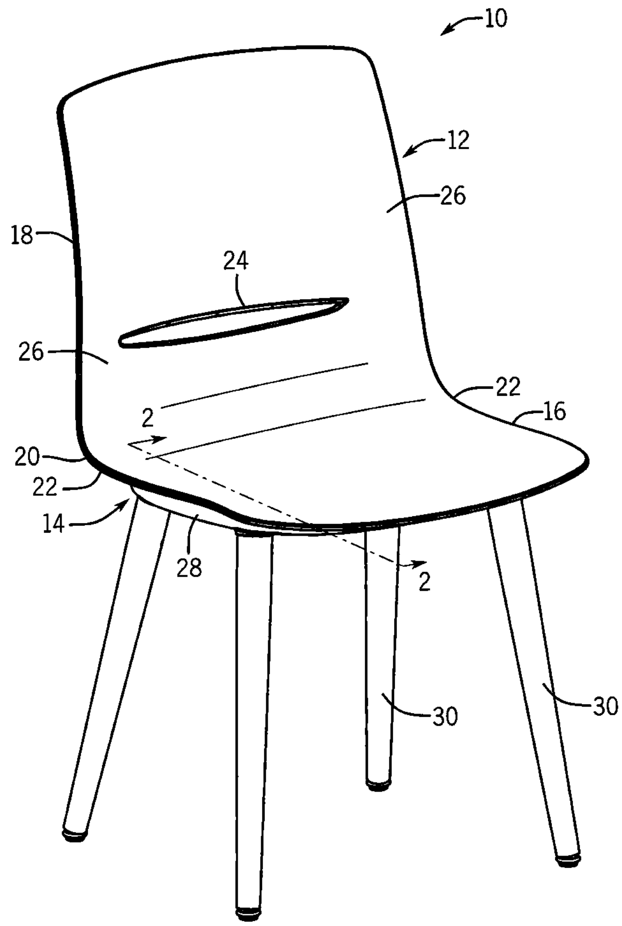

FIG. 1

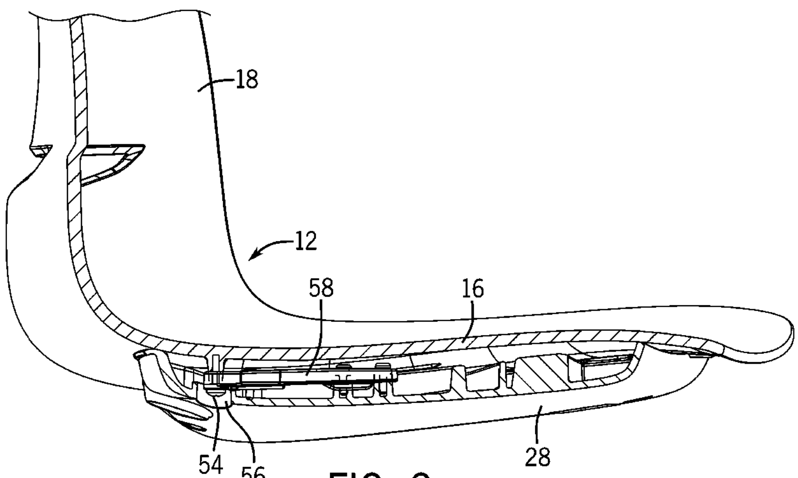

FIG. 2

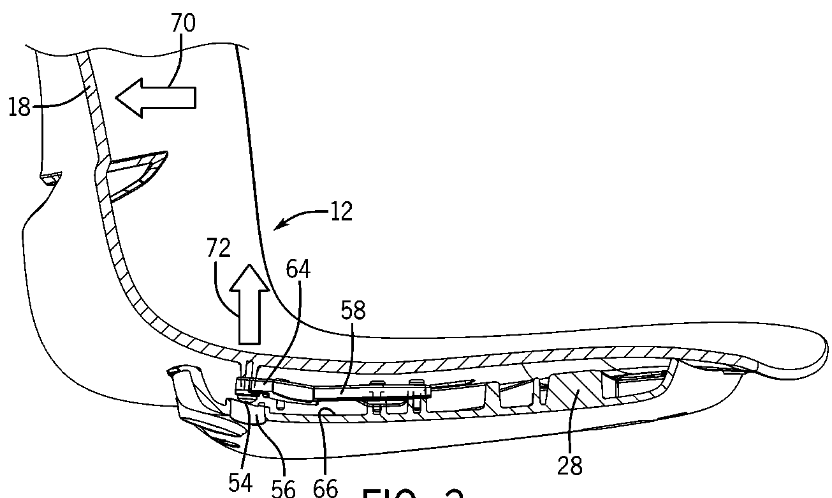

FIG. 3

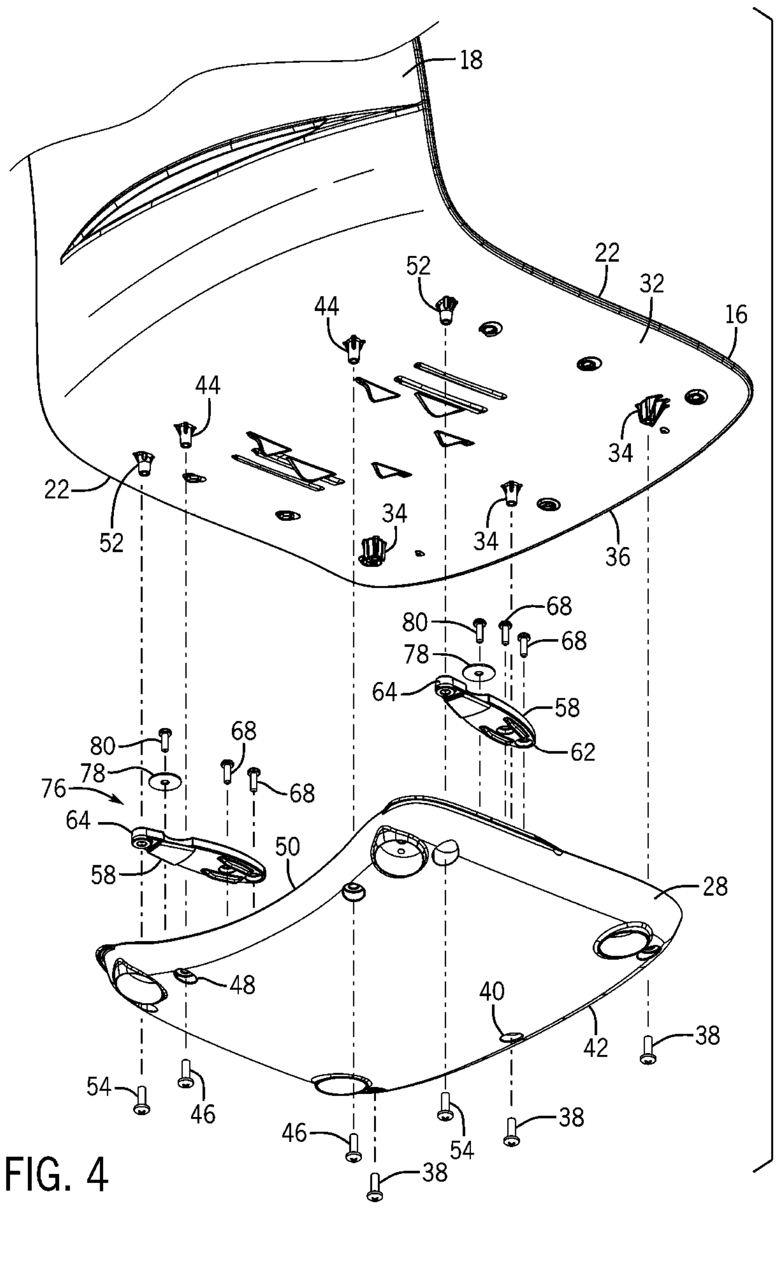
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FIG. 4

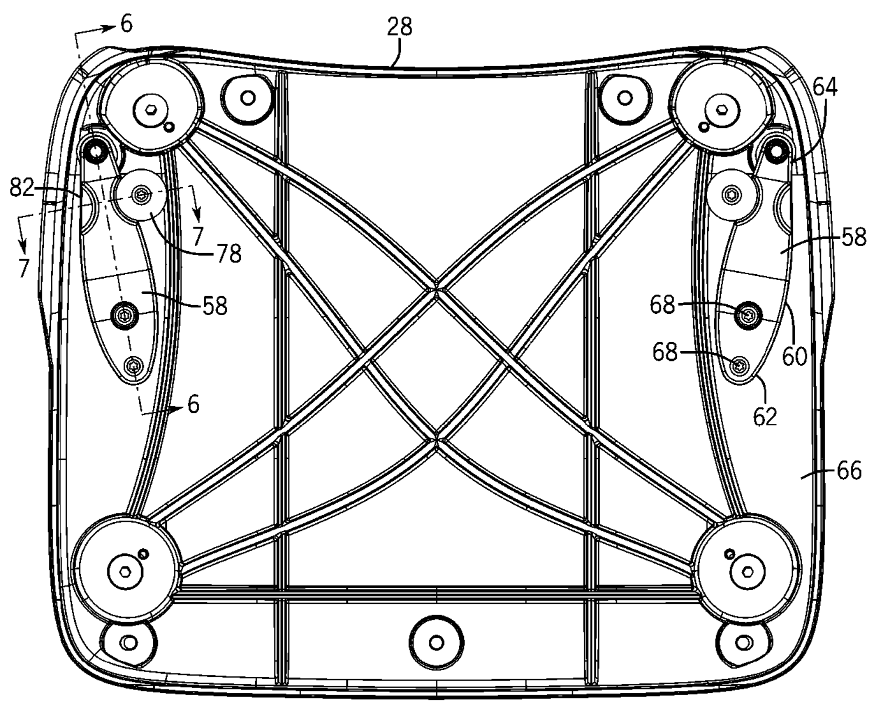

FIG. 5

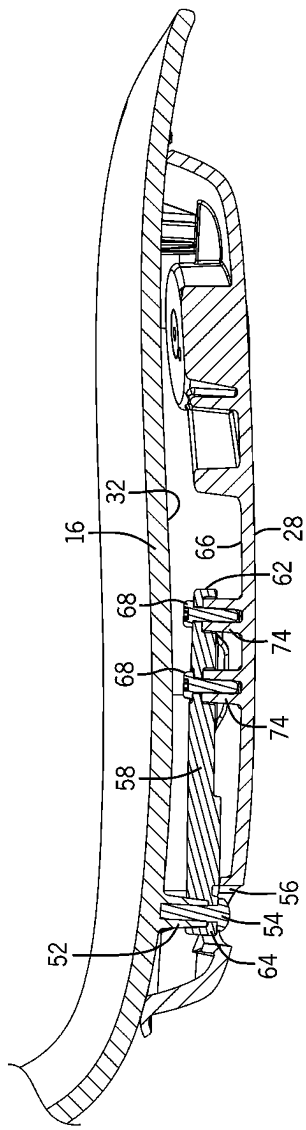

FIG. 6

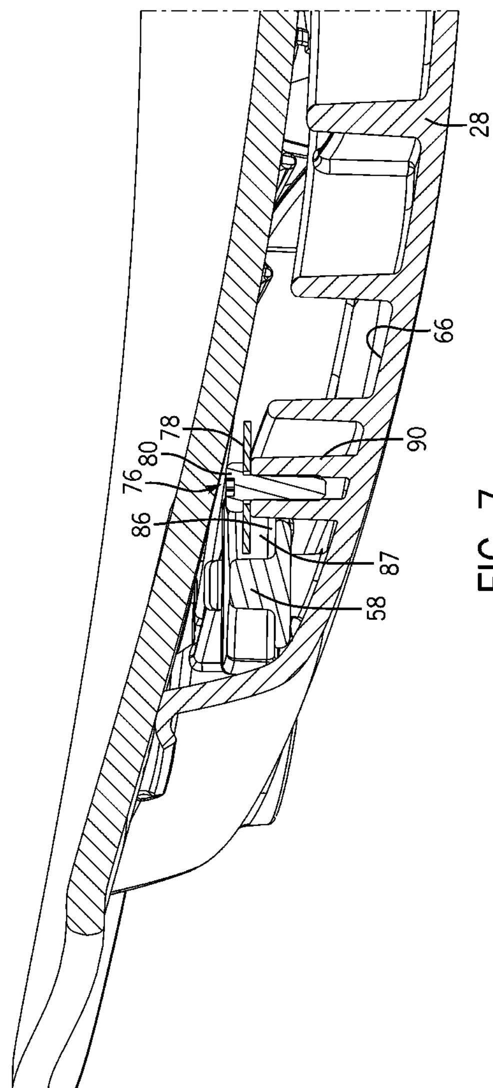

FIG. 7

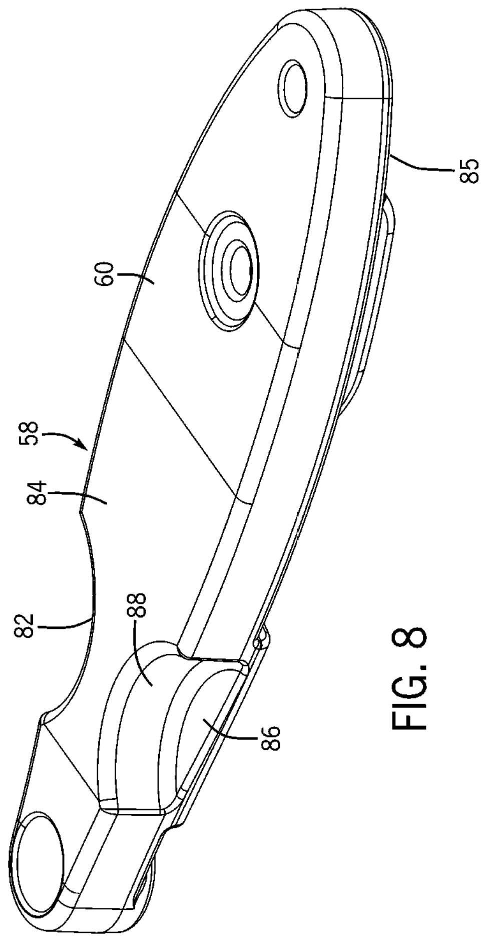

FIG. 8

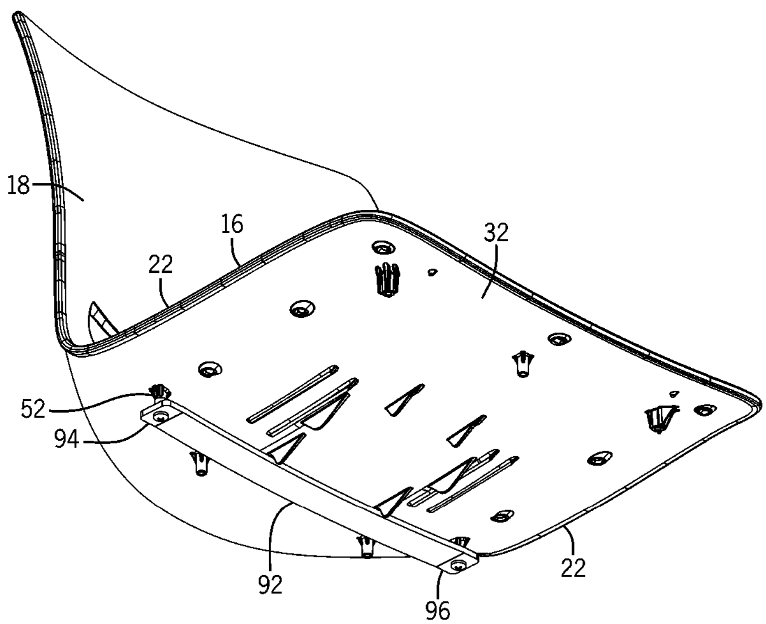

FIG. 9

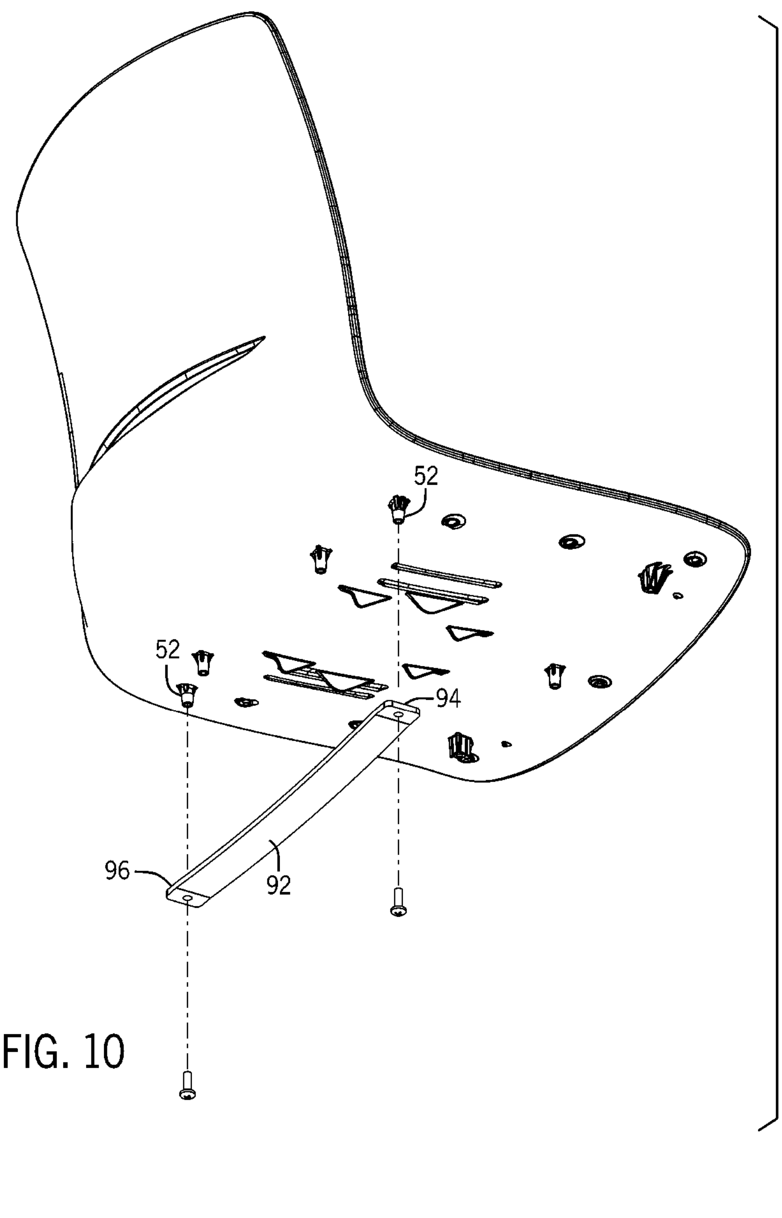

FIG. 10

FLEXING CHAIR SHELL WITH FLEXIBLE ELEMENTS

BACKGROUND

The present disclosure generally relates to a chair having a one-piece molded shell for supporting a user. More specifically, the present disclosure relates to a chair that includes one or more flexible element that limit the inward flexing of the side edges of a seat upon reclining movement of the backrest.

Plastic chairs in which a seat and a backrest are molded as a single shell are well known. In many of these chair designs, the backrest is flexible and is able to pivot and recline relative to the seat. During the reclining movement, the side edges of a transition area between the seat and the backrest along with the seat itself have a tendency to pinch inward toward the seated occupant. The more the backrest reclines, the more the side edges of the seat pinch inward toward the seated occupant.

The inventor of the present disclosure has recognized the need and desire for a chair that includes some type of element or elements that restrict the reclining movement of the backrest. The present disclosure provides one or more flexible elements that restrict the reclining movement of the backrest and create a bias force that help urge the backrest into an upright, resting position.

SUMMARY

The present disclosure relates to a chair having a molded, one-piece shell that is flexible enough to allow a backrest of the chair to recline relative to the seat. More specifically, the present disclosure relates to a chair that includes one or more flexible elements that restrict and control the flexing of the seat and provide a bias force to return the backrest to a resting position.

In accordance with one aspect of the present disclosure, a chair is provided for use by a seat occupant. The chair includes a support structure that includes a base and a plurality of legs connected to the base to support the base above a surface, such as the ground. The chair includes a one-piece molded seat shell that is supported on the base of the support structure. The seat shell includes a seat having a seating surface, a bottom surface and a pair of side edges. In addition, the seat shell includes a backrest that can pivot relative to the seat upon a reclining force applied to the backrest by the seat occupant.

When the seat occupant exerts the reclining force on the backrest, the backrest reclines and the side edges of the seat flex inward and upward. The flexing movement of the side edges is toward the seated occupant. The chair of the present disclosure includes one or more flexible members that are positioned to restrict the upward and inward flexing movement of the side edges of the seat.

In one contemplated, exemplary embodiment, the chair includes a pair of flexible element that are each positioned near one of the side edges of the seat. Each of the pair of flexible elements has a first end connected to the stationary base and a second end connected to the bottom surface of the seat. The pair of flexible elements are the connection between the base and the side edges of the seat.

In an exemplary embodiment of the present disclosure, the chair further includes a stop member positioned near each of the flexible members. The stop member is designed to limit the vertical movement of the flexible member to thereby limit the amount of reclining movement of the backrest. The stop member limits the vertical movement of the flexible member and thus the flexing of the side edges of the seat.

Various other features, objects and advantages of the invention will be made apparent from the following description taken together with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The drawings illustrate the best mode presently contemplated of carrying out the disclosure. In the drawings:

FIG. **1** is a front perspective view of a chair including a one-piece shell and support structure in accordance with the present disclosure;

FIG. **2** is a section view taken along line **2-2** of FIG. **1** showing the resilient mounting arrangement of the present disclosure between the shell and the base;

FIG. **3** is a section view similar to FIG. **2** showing the movement of the side edges of the seat upon reclining movement of the backrest;

FIG. **4** is an exploded view showing the mounting components between the shell and the base;

FIG. **5** is a top view of the base including the pair of resilient flexible members;

FIG. **6** is a section view taken along line **6-6** of FIG. **5** showing the mounting of one of the flexible members to both the seat and the base;

FIG. **7** is a section view taken along line **7-7** of FIG. **5** showing the position of the stop member relative to one of the flexible members;

FIG. **8** is a top perspective view of the flexible member;

FIG. **9** is a bottom perspective view of a second contemplated embodiment of the present disclosure; and

FIG. **10** is an exploded view of the second contemplated embodiment of the present disclosure.

DETAILED DESCRIPTION

FIG. **1** illustrates a chair **10** constructed in accordance with the present disclosure. The chair **10** generally includes a one-piece shell **12** that is supported above a ground surface by a support structure **14**. In the embodiment illustrated in FIG. **1**, the one-piece shell **12** is formed from a molded plastic material such that the one-piece shell **12** defines a seat **16** and a backrest **18** that are joined to each other by a curved transition portion **20**. The plastic material that forms the one-piece shell **12** is flexible enough to allow the backrest **18** to recline upon a reclining force exerted against the backrest **10** by a seated occupant. During this reclining movement, the backrest **18** flexes rearward, which causes the side edges **22** of seat **16** to flex inward toward the center axis of the seat **16**. This inward flexing of the side edges **22** of the seat **16** can impinge upon the seated occupant if the amount of flexing of the backrest **18** is not restricted. The inventor of the present disclosure recognized this need and developed the subject matter of the present disclosure to address this issue.

In the embodiment shown in FIG. **1**, the backrest **18** includes a removed area **24** that creates a pair of flexing zones **26** in the backrest **18** on opposite sides of the removed area **24**. Because of the lateral curves formed in the geometry of the one-piece shell **12**, the flexing of the backrest **18** rearward results in a tightening of the lateral curve of the seat **16** and the transition portion **20** at the hip portion of a seated occupant. In order to restrict this inward movement of the side edges **22** of the seat **16**, the present disclosure utilizes one or more flexible elements to restrict such movement, as will be described in greater detail below.

In the embodiment shown in FIG. **1**, the support structure **14** for the chair **10** includes stationary base **28** that is supported by four legs **30**. The base **28** can be formed from various different types of materials, such as a molded plastic, molded nylon or a cast metallic material. In the embodiment shown, the base **28** is formed from a cast metal material and securely receives each of the four legs **30**.

FIG. **4** is an exploded view of the mounting arrangement between a bottom surface **32** of the seat **16** and the base **28** in accordance with the present disclosure. The mounting arrangement between these two components prevents the lateral movement of the seat **16** relative to the base **28** while also allowing the side edges **22** of the seat to flex both inward and upward upon reclining movement of the backrest **18**.

As illustrated in FIG. **4**, the bottom surface **32** of the seat **16** includes three front standoffs **34** positioned near the front edge **36** of the seat. Each of the front standoffs **34** includes an internally threaded opening that is designed to receive one of the front connectors **38**. Each of the front connectors **38** passes through an attachment hole **40** aligned with the front standoffs **34** and located near the front edge **42** of the base **28**. The threaded shaft of each of the front connectors **38** is received in the internally threaded opening formed in the front standoff **34**. The series of front connectors **38** thus secure the front edge **36** of the seat **16** in a fixed position relative to the front edge **42** of the base **28**. Such connection prevents the lateral movement of the front edge **36** of the seat upon reclining movement initiated by the seat occupant.

The bottom surface **32** of the seat **16** further includes a pair of rear standoffs **44** that each include a threaded inner opening designed to receive a threaded shaft of one of the pair of rear connectors **46**. Each of the rear connectors **46** passes through an attachment hole **48** located near the back edge **50** of the base. The rear connectors **46** are spaced inwardly from the side edges **22** of the seat and are used to secure the center portion of the seat, near the rear edge, to the base **28**.

The bottom surface **32** of the seat **16** further includes a pair of side edge standoffs **52** that are each located near one of the side edges **22** of the seat **16** and near the rear most portion of the seat **16** where the seat **16** transitions into the transition portion **20**. Each of the side edge standoffs **52** include an internal threaded opening that receives a threaded shaft of a side edge connector **54**. Each of the side edge connectors **54** pass through an access opening **56** formed in the base **28**. The access openings **56** are sized such that the side edge connector **54** passes entirely through the base and does not engage any portion of the base **28**. In this manner, the side edges **22** of the seat are able to flex both upward and inward upon reclining movement of the backrest **18**.

Referring now to FIGS. **4** and **5**, the chair constructed in accordance with the present disclosure is designed to include a pair of resilient flexible members **58** that are mounted near the side edges **22** of the seat and are connected between the base **28** and the seat **16**. As illustrated in FIG. **5**, each of the flexible members **58** includes a main body **60** that extends from a first end **62** to a second end **64**. Although a specific shape of the main body **60** is shown, it should be understood that the shape of the main body **60** could vary depending on the requirements for the flexible member **58**. The first end **62** of the flexible member **58** is securely connected to an inner surface **66** of the base **28** using a pair of connectors **68**. The pair of connectors **68** securely mount the first end **62** of the flexible member **58** to the base **28** such that the first end **62** is prevented from movement relative to the base **28**.

The second end **64** of the flexible member **58** is securely mounted to one of the side edge standoffs **52** on the bottom surface **32** of the seat **16** utilizing one of the side edge connectors **54**. As indicated previously, each of the side edge connectors **54** pass through the access opening **56** formed in the base **28** such that the rear portion of the seat **16** near the side edges **22** are not securely attached to the stationary base **28**. Instead, the side edges **22** are able to flex upward and inward against a resilient spring-like force created by the material used of the flexible members **58**.

Referring back to FIGS. **2** and **3**, when a seat occupant is seated on the seat and leans back against the backrest **18** as shown by arrow **70**, the plastic material used to form the one-piece shell **12** allows the side edges of the seat **16** to flex upward and inward as shown by arrow **72**. As the side edge **22** flexes upward and inward, the body of the flexible member **58** secured to the seat also flexes upward, which allows the second end **64** to move upward away from the inner surface **66** of the base **28**. As shown in FIG. **3**, the access opening **56** allows the side edge connector **54** and the second end **64** of the flexible member **58** to move upward. When the seat occupant no longer exerts a reclining force against the backrest **18**, the resilient nature of the flexible member **58** urges the backrest **18** back into the resting position shown in FIG. **2**. In this manner, the flexible member **58** not only allows the flexing movement of the side edges of the seat, but also exerts a bias force on the side edges of the seat to urge the backrest back into the upright resting position shown in FIG. **2**.

As shown in FIGS. **2** and **3**, the flexible members **58** are positioned between the seat **16** and the base **28** and are thus concealed within the chair. Hiding the flexible members **58** within the chair enhances the visual appearance of the chair while allowing for the function described above.

In the embodiment illustrated, each of the flexible members **58** can be formed from a flexible material having the desired thickness to exert the required return force on the backrest **18** while yet allowing the required flexing of the side edges of the seat inward upon the inclining movement.

FIG. **6** more clearly illustrates the mounting of the flexible member **58** between the base **28** and the bottom surface **32** of the seat **16**. As illustrated, the first end **62** of the flexible member **58** is securely mounted to the base by a pair of connectors **68** that are each received in one of a pair of standoffs **74** extending upward from the inner surface **66** of the base **28**. The pair of connectors **68** thus secure the first end **62** of the flexible member **58** to the base **28**. The second end **64** of the flexible member **58** is securely connected to the bottom surface **32** of the seat **16** through the side edge connector **54**, which is received in the side edge standoff **52**. The size of the side edge connector **54** is less than the diameter of the access opening **56** formed in the base **28** such that the second end of the flexible member **58** is movable vertically as illustrated.

Referring back to FIGS. **4** and **5**, the flexible mounting system of the present disclosure further includes a pair of stop members **76** that are each associated with one of the pair of flexible members **58**. The stop member **76** is designed to limit the upward movement of the second end **64** of the flexible member **58**, which is illustrated by the arrow **72** in FIG. **3**. The stop member is positioned and located relative to the flexible member **58** to provide a definitive stop position for the upward movement of the second end **64** of the flexible member **58**. In the embodiment illustrated, the stop member **76** includes a washer **78** and a connector **80**. As can be seen in FIG. **5**, the washer **78** has a portion of its outer circumference that extends over a recessed area **82** formed in the flexible member **58**.

Referring now to FIG. **8**, in the embodiment illustrated, the body **60** of the flexible member includes a top surface **84** and a bottom surface **85** space by the thickness of the body **60**. The body **60** includes the pair of recessed areas **82** that each extend from the top surface **84** to a contact surface **86**. The shape of the recessed area **82** is defined by an inner wall **88** that has a general shape that corresponds to the outer surface of the washer.

Referring now to FIG. **7**, when the flexible member is in the initial resting position, the washer **78** is spaced above the contact surface **86** to define a movement gap **87**. In the embodiment shown, this movement gap **87** has a height of approximately 0.25 inches, although other heights are contemplated. When a seat occupant exerts a reclining force on the backrest, the second end of the flexible member **58** moves upward until the contact surface **86** comes in contact with the washer **78**. This physical contact between the washer **78** and the contact surface **86** formed on the flexible member prevents any additional upward movement of the second end of the flexible member. In this manner, the stop member **76**, and specifically the stationary washer **78**, limits the amount of movement of the flexible member **58** and thus the side edges of the seat. As shown in FIG. **7**, the threaded shaft of the connector **80** is received within a standoff **90** extending upward from the inner surface **66** of the base **28**.

As illustrated in FIGS. **4** and **5**, each of the flexible members **58** includes a stop member **76** to limit the amount of movement allowed for the second end of the flexible member. The stop member **76** could have other shapes and configurations as long as the stop member **76** limits the amount of movement of the flexible member **58**. In some embodiments, the stop member **76** could be eliminated if some other mechanism is present to limit the movement of the backrest.

As described previously, each of the flexible members **58** can be formed from a molded material that has the desired flexibility and that creates the desired spring force to return the backrest to its upright position shown in FIG. **2**. Although one shape of the flexible member **58** is shown in FIG. **8**, it should be understood that various other shapes could be utilized while operating within the scope of the present disclosure.

FIGS. **9** and **10** illustrate a second embodiment of the flexible mounting arrangement in accordance with the present disclosure. As can be understood in FIGS. **9** and **10**, the bottom surface **32** of the seat **16** includes the same standoffs and general configuration as in the embodiment shown in FIG. **4**. In the second embodiment shown in FIG. **10**, the pair of flexible members **58** previously described are replaced with a single flexible element **92** that extends across the general width of the seat **16** between the pair of side edge standoffs **52**. In the embodiment illustrated, the flexible member **92** functions similar to a leaf spring and has a first end **94** and a second end **96** of the flexible member **92** securely attached to the bottom surface **32** of the seat.

In the embodiment shown in FIGS. **9** and **10**, when the seat occupant exerts a reclining force on the backrest **18**, the side edges **22** of the seat **16** begin to move upward as previously discussed. This upward movement of the seat edges **22** is opposed by the flexible nature of the flexible member **92**. Further, when the seat occupant no longer exerts a reclining force on the backrest **18**, the flexible member **92** urges the side edges **22** of the seat back to the resting position shown in FIG. **1**.

In the embodiment shown in the present disclosure, flexible members **58** can be formed from different types of materials or composite combinations of various materials. As an example, the flexible elements could be steel, fiberglass, nylon, rubber or some type of plastic material. The type of material and size and configuration of the flexible member can be designed depending upon the required bias force and the amount of flexing required in a certain chair design.

This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to make and use the invention. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

I claim:

1. A chair for use by a seat occupant, comprising:

a support structure including a base and a plurality of legs connected to the base;

a one-piece molded seat shell supported on the support structure, the seat shell including a seat and a backrest joined to each other by a transition portion, wherein the backrest pivots in relation to the seat upon reclining movement by the seat occupant, wherein side edges of the seat flex inward upon pivoting movement of the backrest; and

at least one resilient flexible member positioned to restrict the flexing movement of the side edges of the seat when the backrest pivots; and

a stop member mounted to the base;

wherein the stop member is spaced apart from the flexible member and contacts the flexible member to limit the vertical movement of the flexible member.

2. The chair of claim **1** wherein the at least one flexible member restricts the vertical movement of the side edges of the seat upon reclining movement of the backrest.

3. The chair of claim **1** wherein the at least one flexible member includes a first end connected to the base and a second end connected to the seat.

4. The chair of claim **1** wherein the stop member contacts the flexible member upon inward flexing movement of the side edges of the seat.

5. A chair for use by a seat occupant, comprising:

a support structure including a base and a plurality of legs connected to the base;

a one-piece molded seat shell supported on the support structure, the seat shell including a seat having a seating surface, a bottom surface and a pair of side edges and a backrest joined to the seat by a transition portion, wherein when the backrest pivots in relation to the seat upon reclining movement by the seat occupant, the side edges of the seat flex inward; and

a pair of resilient flexible members positioned to restrict the inward flexing movement of the side edges of the seat when the backrest pivots; and

a pair of stop members each mounted to the base;

wherein the pair of stop members are each spaced from one of the pair of flexible members such that the flexible members contact the stop members to limit vertical movement of one of the pair of flexible members.

6. The chair of claim **5** wherein each of the flexible members restricts the vertical movement of the side edges of the seat upon reclining movement of the backrest.

7. The chair of claim **5** wherein each of the flexible members includes a first end connected to the base and a second end connected to the bottom surface of the seat.

8. The chair of claim **7** wherein each of the flexible members is located adjacent to one of the side edges of the seat.

9. The chair of claim **7** wherein each of the flexible members is formed from a solid flexible material such that the second end is movable relative to the first end.

10. The chair of claim **5** wherein the flexible members contact the stop members upon inward flexing movement of the side edges of the seat.

11. The chair of claim **5** wherein the seat is secured to the base to prevent movement of the seat along the base such that pivoting movement of the backrest creates the inward movement of the side edges of the seat.

12. A chair for use by a seat occupant, comprising:
- a support structure including a base and a plurality of legs connected to the base;
- a one-piece molded seat shell supported on the support structure, the seat shell including a seat having a seating surface, a bottom surface and a pair of side edges and a backrest joined to the seat by a transition portion, wherein when the backrest pivots in relation to the seat upon reclining movement by the seat occupant, the side edges of the seat flex inward and upward;

- a pair of resilient flexible members positioned to restrict the inward and upward flexing movement of the side edges of the seat when the backrest pivots; and
- a pair of stop members each mounted to the base and spaced from one of the pair of flexible members;
- wherein the pair of stop members are configured to contact one of the pair of flexible members to limit the vertical movement of one of the pair of flexible members.

13. The chair of claim **12** wherein each of the flexible members includes a first end connected to the base and a second end connected to the bottom surface of the seat.

14. The chair of claim **12** wherein the flexible members contact the stop members upon inward flexing movement of the side edges of the seat.

15. The chair of claim **12** wherein the seat is secured to the base to prevent movement of the seat along the base such that pivoting movement of the backrest creates the inward movement of the side edges of the seat.

16. The chair of claim **12** wherein each of the flexible members is located adjacent to one of the side edges of the seat.

17. The chair of claim **12** wherein each of the flexible members includes a top surface and a contact surface recessed from the top surface, wherein the stop member contacts the contact surface to limit the movement of the flexible member.

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